

---

# POSITIONAL SELF-DISTILLATION FOR LARGE LANGUAGE MODELS

---

A PREPRINT

Astarag Mohapatra  
astarag2843@gmail.com

May 31, 2026

## ABSTRACT

Knowledge distillation for large reasoning models typically applies a uniform loss over every generated token. In this paper, we ask a simple question: does the *position* of a token within a reasoning chain affect how useful it is as a distillation target? We build on On-Policy Self-Distillation (OPSD) [1], which trains a student model by minimizing a forward KL divergence against a fixed teacher, and introduce four position-selective variants that restrict distillation to all tokens within the first, middle, or last sentence of each reasoning paragraph, or to the first token of each paragraph. To motivate this, we run a Mann-Whitney U test on per-token KL proxy values and find that sentence-start tokens exhibit significantly higher variance than middle or end tokens, suggesting they carry richer gradient signal. Our preliminary experiments on Qwen3-1.7B across AIME24, AIME25, and HMMT25 show mixed but occasionally promising results: `parafirsttok_topk1` reaches 40.8% Avg@8 on AIME25 at step 100, compared to the full-token OPSD baseline of 37.9% at that step. These gains are not stable across all checkpoints and benchmarks, and training curves are noisy — we do not claim a definitive improvement. Rather, we offer this as a useful datapoint: a sparser distillation signal concentrated at structurally motivated sentence positions can sometimes match or briefly exceed full-token distillation, which motivates further investigation.

## 1 Introduction

*“The scariest moment is always just before you start. After that, things can only get better.”*  
— Stephen King

Training reasoning models is expensive. Methods like GRPO [4] and self-distillation [1, 3, 2] have shown that a student model can be improved by learning from a stronger (or fixed) teacher’s output distribution over long chain-of-thought completions. However, most of these methods apply the distillation loss uniformly across every token in the generated sequence — including filler words, punctuation, and repetitive connectives that likely carry little information.

Recent work has shown that this uniformity is not necessary. Wang et al. [9] find that roughly 20% of tokens — those with the highest entropy under the model’s distribution — account for the vast majority of learning signal in reinforcement learning for reasoning. These high-entropy “minority tokens” are the decision points: the places where the model is genuinely uncertain, and where getting the prediction right or wrong has the most downstream consequence. This motivates a natural design choice in distillation: focus the training signal on high-entropy tokens and ignore the rest.

The question we explore in this paper is a step further: *where do these high-entropy tokens tend to appear in a reasoning chain?* In long-form mathematical reasoning, a model’s output is structured into paragraphs — each paragraph represents a coherent sub-step or shift in reasoning strategy, and a new paragraph begins when the model transitions to a new line of thought. Within a paragraph, we expect the first sentence to carry more uncertainty than the middle or the end, which tend to elaborate on an already-committed direction. Paragraph boundaries themselves, however, turn out to be a more nuanced story: the very first token of a new paragraph (typically a newline or function word) is nearly deterministic, and our analysis reveals it is in fact the *lowest*-entropy position (Section 3).

We test this hypothesis empirically. Using a per-token KL proxy, we find that sentence-start tokens have significantly higher variance than middle or end tokens (Section 3). We then introduce four position-selective distillation variants on top of OPSD and evaluate them on three mathematical reasoning benchmarks (Section 5).

Our contributions are:

- A positional entropy analysis showing that sentence-start tokens have significantly higher KL proxy variance than middle or end tokens (Section 3).
- Four position-selective distillation variants that restrict training signal to specific sentence positions within paragraphs (Section 4).
- Preliminary experimental results on three math benchmarks, with an honest discussion of what works, what does not, and why the results are inconclusive (Section 6).

## 2 Background

**On-Policy Self-Distillation (OPSD).** OPSD [1] trains a student model on reasoning traces sampled from itself (on-policy), using a fixed copy of the model as the teacher. The training objective minimizes a clipped forward KL divergence between the teacher’s token-level distribution  $p_{\text{teacher}}$  and the student’s  $p_{\text{student}}$ :

$$\mathcal{L}_{\text{OPSD}} = \sum_t \text{clip}(\log p_{\text{teacher}}(x_t) - \log p_{\text{student}}(x_t), -c, c) \quad (1)$$

where  $c$  is a clipping threshold (set to 0.05 in our experiments). The student is a LoRA adapter [5] on top of a pretrained base model, and the teacher is a frozen copy of the initial model. Training data comes from OpenThoughts [6], a dataset of long-form mathematical reasoning traces.

**Selective distillation.** Prior work on knowledge distillation has explored selecting which tokens or layers to distill. Methods like MiniLLM [7] and DistiLLM [8] focus on reverse KL or sequence-level objectives. Positional selection within generated reasoning chains — restricting distillation to structurally motivated token positions — has received less attention.

## 3 Motivation: Positional Entropy Analysis

Before designing position-selective variants, we ask whether different sentence positions in a reasoning paragraph have measurably different distillation signal. We use the per-token KL proxy  $\log p_{\text{teacher}} - \log p_{\text{student}}$  as a proxy for how much each token contributes to the distillation loss.

We classify each token in a set of OpenThoughts reasoning traces by its sentence position within its paragraph: *first sentence*, *middle sentence*, *last sentence*, or *single-sentence paragraph*. Table 1 reports the mean and standard deviation of this proxy, along with 95% confidence intervals.

Table 1: Per-token KL proxy ( $\log p_{\text{teacher}} - \log p_{\text{student}}$ ) by sentence position.

| Position | Mean   | Std   | 95% CI           | $n$       |
|----------|--------|-------|------------------|-----------|
| Start    | −0.188 | 1.713 | [−0.193, −0.183] | 535,999   |
| Middle   | −0.083 | 0.567 | [−0.085, −0.080] | 212,273   |
| End      | −0.100 | 0.553 | [−0.101, −0.098] | 497,750   |
| Single   | −0.118 | 1.220 | [−0.119, −0.117] | 8,871,007 |

The start position has notably higher standard deviation (1.617) compared to middle (0.493) and end (0.569), indicating that sentence-start tokens are more spread out in terms of how much the teacher and student disagree. We run one-sided Mann-Whitney U tests to check whether start tokens have higher KL proxy values than other positions:

While the means are all negative (student slightly overconfident relative to teacher), the variance at the start position is much larger (std = 1.713 vs. 0.567 for middle and 0.553 for end), meaning the signal is more extreme in both directions. We hypothesize that these high-variance tokens represent decision points where the student’s predictions diverge most from the teacher, making them more informative training targets.

Table 2: One-sided Mann-Whitney U test results (row &gt; column).

| Comparison     | $p$ -value             | Interpretation                                    |
|----------------|------------------------|---------------------------------------------------|
| Start > Middle | 1.00                   | n.s. — start has higher variance, not higher mean |
| Start > End    | $2.43 \times 10^{-59}$ | significant                                       |
| Middle > End   | 0.00                   | highly significant                                |

**A caveat on paragraph-first tokens.** We also separately analyzed the first token of each paragraph (token-0 of the first sentence), since paragraph boundaries are natural structural anchors. Counter-intuitively, these tokens are the *lowest*-entropy position: mean KL proxy =  $-0.967$  (std =  $5.002$ ,  $n = 461,676$ ), versus  $-0.078$  for the rest of first-sentence tokens. A Mann-Whitney test confirms that token-0 is stochastically *lower* than the remaining tokens ( $p = 1.00$  for the “tok-0 > rest” alternative). This makes sense in retrospect: the first token of a new paragraph is typically a formatting or function word (“\n”, “The”, “So”) whose identity is nearly deterministic given context, so the student is extremely confident and contributes almost no loss. The entropy is concentrated in the body of the first sentence, not its opening token. This foreshadows the underwhelming performance of `parafirsttok_topk1` in our experiments (Section 6).

## 4 Method: Position-Selective Token Distillation

**Paragraph and sentence segmentation.** We segment each generated reasoning trace into paragraphs by splitting on blank lines. Within each paragraph, we identify sentences using punctuation-based heuristics. For a paragraph with  $S$  sentences, we label sentence 1 as *first*, sentence  $S$  as *last*, and sentences  $2, \dots, S - 1$  as *middle*. A paragraph with a single sentence is treated as both first and last.

**Token selection.** For each selected sentence position, all tokens within that sentence contribute to the distillation loss. All tokens outside the selected position receive zero loss weight (implemented by setting their labels to  $-100$ ).

We test the following four variants, plus the full-token OPSD baseline:

1. **firstsent\_topk1**: all tokens in the first sentence of each paragraph
2. **middlesent\_topk1**: all tokens in the middle sentence of each paragraph
3. **lastsent\_topk1**: all tokens in the last sentence of each paragraph
4. **parafirsttok\_topk1**: only the single first token of each paragraph

## 5 Experiments

**Setup.** We fine-tune Qwen3-1.7B [10] using LoRA ( $r = 64$ ,  $\alpha = 128$ , applied to all projection matrices) on OpenThoughts data. Training uses AdamW with learning rate  $5 \times 10^{-6}$ , gradient clipping at 0.1, temperature 1.1, top- $p = 0.95$ , top- $k = 20$ , and  $\beta = 0$  (no KL penalty on the policy). The teacher is a fixed copy of the initial model. We train for 100 steps and evaluate checkpoints at steps 25, 50, 75, and 100.

**Evaluation.** We evaluate on three mathematical reasoning benchmarks: AIME 2024 (AIME24), AIME 2025 (AIME25), and HMMT 2025 (HMMT25). Inference uses vLLM with thinking mode enabled, temperature 1.0, top- $p = 0.95$ , top- $k$  disabled, min- $p = 0.0$ , presence penalty 0.0, and a maximum of 38,912 new tokens. We draw  $n = 8$  samples per problem and report three metrics: **Pass@8** (fraction of problems solved by at least one sample), **MajVote@8** (majority-vote accuracy over 8 samples), and **Avg@8** (mean per-sample accuracy).

**Benchmark results.** Figures 1, 2, and 3 show all metrics across training steps for all variants.

## 6 Results and Discussion

The honest summary is that results are noisy and no single variant consistently outperforms the full-token OPSD baseline across all benchmarks and metrics. However, a few patterns are worth noting.

**First and last sentence variants occasionally match the baseline.** `firstsent_topk1` and `lastsent_topk1` occasionally match or slightly exceed the baseline on AIME24 and AIME25 at certain checkpoints. This is notable

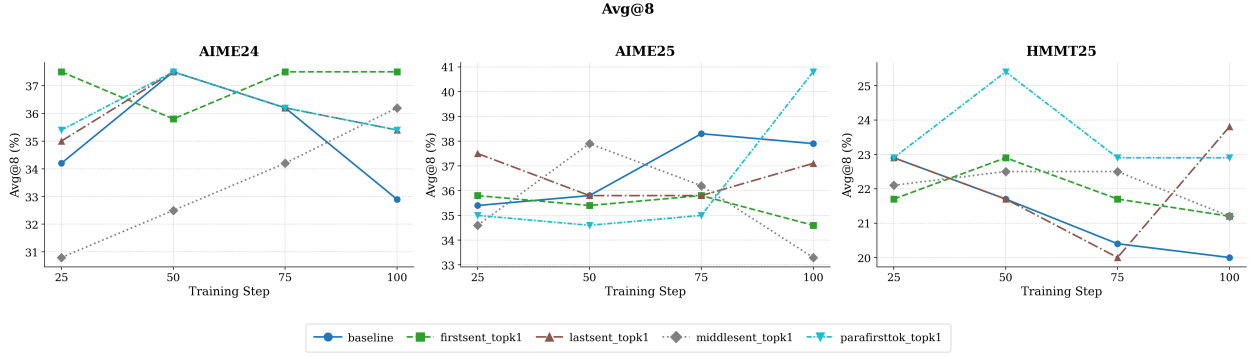


Figure 1: Avg@8 across training steps on AIME24, AIME25, and HMMT25.

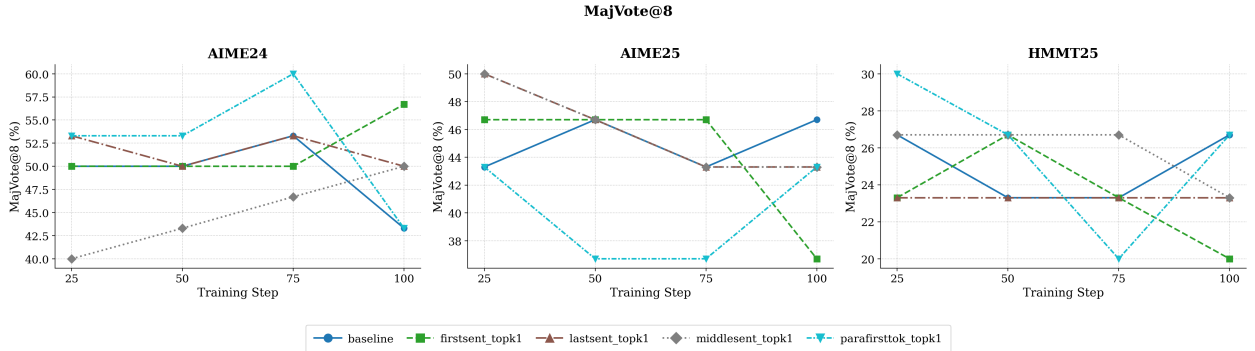


Figure 2: MajVote@8 across training steps on AIME24, AIME25, and HMMT25.

because these variants restrict distillation to a single sentence per paragraph, achieving similar performance with a substantially sparser training signal.

**Middle sentence underperforms.** `middlesent_topk1` is generally the weakest variant across all three benchmarks, which is consistent with the lower-entropy observation in Section 3 — middle-sentence tokens have the lowest KL proxy variance and may carry the least informative signal.

**Paragraph-first token is a low-entropy anchor.** `parafirsttok_topk1` shows variable behavior across benchmarks. The entropy analysis in Section 3 explains why: the first token of a paragraph (typically a newline, “The”, or “So”) has a mean KL proxy of  $-0.967$ , far more negative than any other group, meaning the student is already highly confident there. Restricting distillation to these near-zero-loss tokens provides very little gradient signal, which likely explains the inconsistent and generally unimpressive results for this variant.

**Training curves are unstable.** All variants show significant fluctuation across the four checkpoints, which makes it difficult to draw firm conclusions from only 100 training steps. A longer run with more checkpoints and multiple seeds would be needed to say anything definitive.

**Limitations.** This is a small exploratory study. We only train one model size (1.7B parameters), run for 100 steps without a hyperparameter search, and evaluate on a narrow set of math benchmarks. The sentence-level selection granularity is coarse — a natural next step is to select individual high-KL tokens within the chosen sentence rather than all tokens. We also did not explore combining positions (e.g., first and last sentence together).

## 7 Conclusion

We asked whether the position of tokens in a reasoning chain affects their value as distillation targets. The entropy analysis in Section 3 provides a principled reason to suspect that sentence-start tokens are more informative, and the preliminary results in Section 6 do not contradict this — but they are also far from conclusive.

The most interesting observation is that sparse, position-selective distillation can sometimes match full-token distillation. If this holds up in longer training runs, it would suggest that most of the tokens in a reasoning chain contribute little

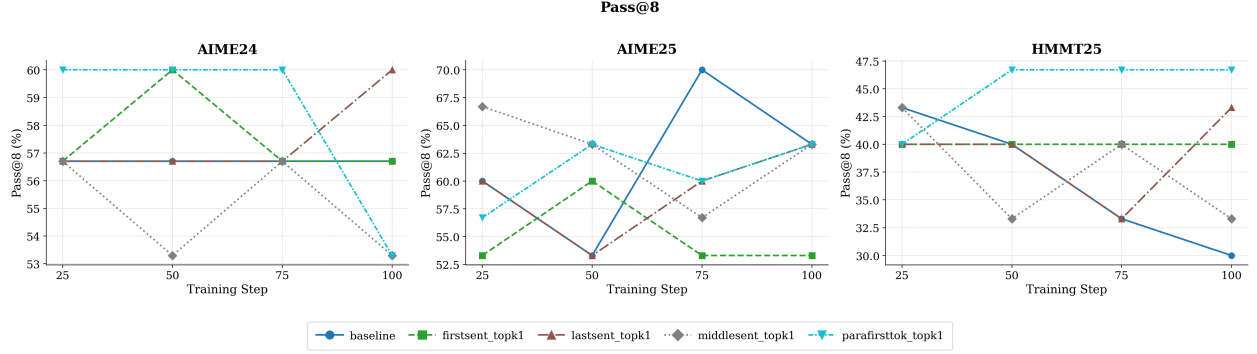


Figure 3: Pass@8 across training steps on AIME24, AIME25, and HMMT25.

to the distillation objective, and that a smarter selection strategy could substantially reduce the computational cost of distillation.

We see this paper as a small step toward understanding which parts of a reasoning chain matter most for training. Future work should explore longer training, multiple seeds, larger model sizes, adaptive position selection, and combinations of positions.

## References

- [1] Siyan Zhao, Zhihui Xie, Mengchen Liu, Jing Huang, Guan Pang, Feiyu Chen, and Aditya Grover. Self-Distilled Reasoner: On-Policy Self-Distillation for Large Language Models. *arXiv preprint arXiv:2601.18734*, 2026.
- [2] Idan Shenfeld, Mehul Damani, Jonas Hübner, and Pulkit Agrawal. Self-distillation enables continual learning. *arXiv preprint arXiv:2601.19897*, 2026.
- [3] Jonas Hübner, Frederike Lübeck, Lejs Behric, Anton Baumann, Marco Bagatella, Daniel Marta, et al. Reinforcement Learning via Self-Distillation. *arXiv preprint arXiv:2601.20802*, 2026.
- [4] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- [5] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *ICLR*, 2022.
- [6] OpenThoughts Team. OpenThoughts: A dataset of long-form reasoning traces. <https://huggingface.co/datasets/open-thoughts/OpenThoughts-114k>, 2025.
- [7] Yuxian Gu, Li Dong, Furu Wei, and Minlie Huang. MiniLLM: Knowledge distillation of large language models. In *ICLR*, 2024.
- [8] Jongwoo Ko, Sungnyun Kim, Tianyi Chen, and Se-Young Yun. DistiLLM: Towards streamlined distillation for large language models. In *ICML*, 2024.
- [9] Shenzhi Wang, Le Yu, Chang Gao, Chujie Zheng, Shixuan Liu, Rui Lu, et al. Beyond the 80/20 rule: High-entropy minority tokens drive effective reinforcement learning for LLM reasoning. In *Advances in Neural Information Processing Systems*, volume 38, pages 115452–115486, 2025.
- [10] Qwen Team. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.